

Multiple Choice (10 marks)

1. The lunar maria are:
 - (a) ancient heavily cratered highlands
 - (b) dark lavas inside volcanic calderas
 - (c) dark lavas filling older impact basins
 - (d) the bright regions on the Moon
2. What effect or effects would be most significant if the Moon's orbital plane were exactly the same as the ecliptic plane?
 - (a) Solar eclipses would be much rarer.
 - (b) Solar eclipses would last much longer.
 - (c) Solar eclipses would be much more frequent.
 - (d) Solar eclipses would not last as long.
3. How do we know how old the Earth is?
 - (a) From the layering of materials within the Earth.
 - (b) From fossils of ancient life.
 - (c) From the cratering history of Earth's surface.
 - (d) From radioactive dating of rocks and meteorites.
4. The constellation ... is a bright W-shaped constellation in the northern sky.
 - (a) Centaurus
 - (b) Cygnus
 - (c) Cassiopeia
 - (d) Cepheus
5. Why is the Mars Exploration Rover Spirit currently tilted towards the north?
 - (a) Because it's climbing up a big hill.
 - (b) Because it's in the southern hemisphere where it is winter now.
 - (c) Because it's in the northern hemisphere where it is winter now.
 - (d) Because one of its wheels broke.

True / False (10 marks)

Answer True or False

6. True or False: The Moon's rotational and orbital periods are equal, resulting in the same face being visible from Earth.
7. True or False: Meteorites with high metal content are typically pieces of comets rather than from asteroids.
8. True or False: A neutral atom always contains equal numbers of neutrons and protons, resulting in a standard atomic mass for all elements.
9. True or False: Galileo invented the telescope in 1609, significantly advancing the field of observational astronomy.
10. True or False: Meteorites typically exhibit extensive processing due to their formation in space.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the significance of Barnard's Star as the second closest star system to Earth, including its characteristics, discovery, and implications for astrophysical research.
12. Discuss the rationale behind Ptolemy's use of 'circles upon circles' in his geocentric model, particularly in relation to the observed retrograde motion of planets.

End of Paper